Project Part 3: Statistical Modeling

Graduate students only (BIOL 5860)

The third of four checkpoint deliverables for the graduate project. Render this file to PDF and submit both the `.qmd` and the rendered `.pdf` via Canvas by the end of Week 13. See the syllabus for the late policy.

1. Model

Fit the statistical model or test that matches your design (revisit your plan from Project Part 1 — update it here if your thinking has changed, and say why).

2. Assumptions and diagnostics

Check the assumptions relevant to your model (residual diagnostics, equal variance, overdispersion, or whatever applies) the way we've practiced throughout the course.

Do the diagnostics look acceptable? If not, what did you do about it (transformation, a different model, a nonparametric alternative)?

3. Results

Report your results in a results-section style summary: effect sizes, test statistics, degrees of freedom, p-values, and confidence intervals where relevant.

4. Interpretation

In 3–5 sentences, interpret your results in the context of your original research question from Part 1. What do they mean biologically, and do they support or refute your hypotheses?